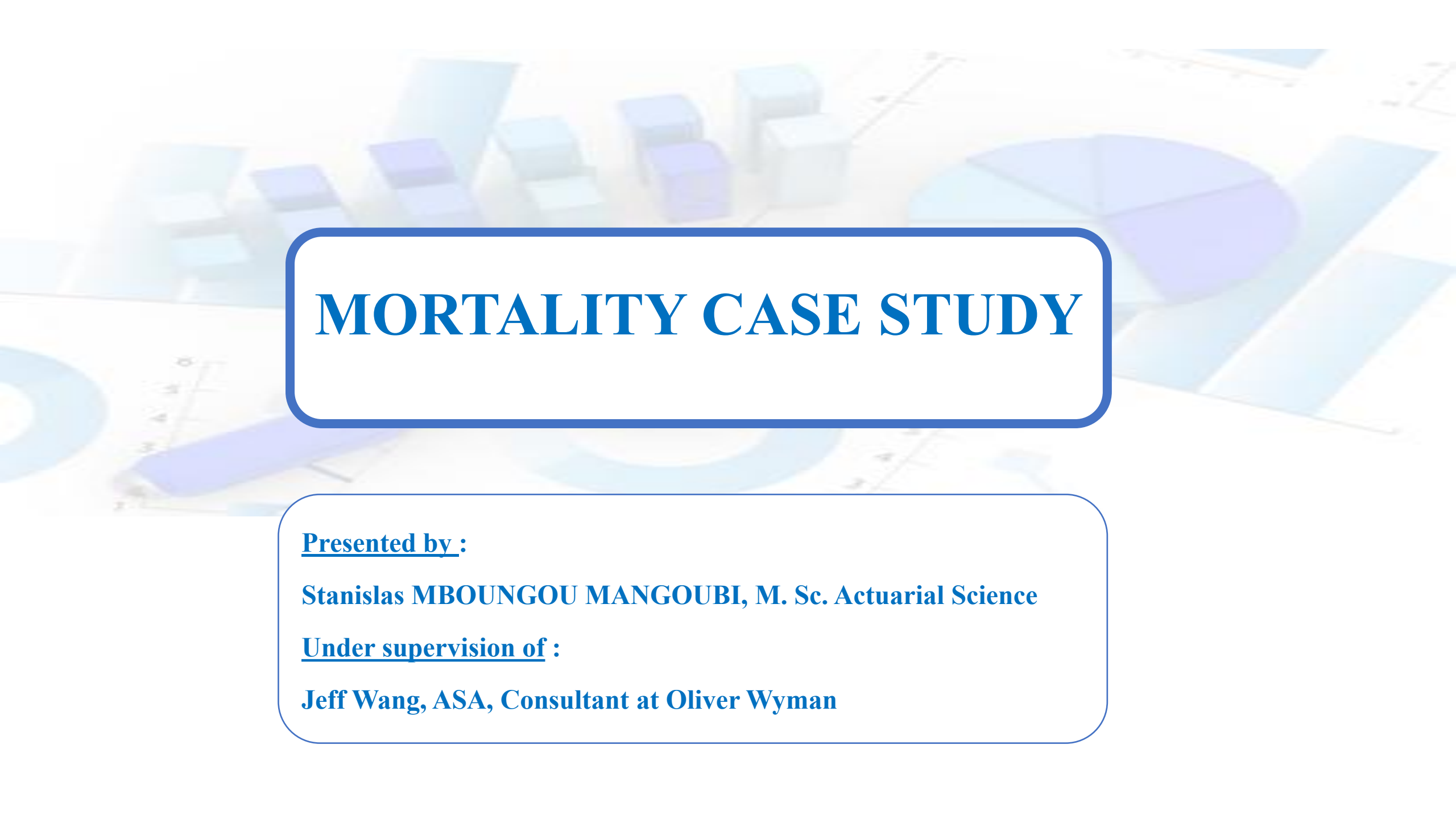




MORTALITY CASE STUDY

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PLAN



In general, the mortality rates of a portfolio of policyholders vary due to changes in experience of subpopulations and shifts in the business mix.

The main objective of this analysis is to systematically explore the client's policyholder data and identify:

- Changes in mortality rates within key subpopulations
- Shifts in the composition of the portfolio that may contribute to increased overall mortality.

MORTALITY ANALYSIS

Evolution of Portfolio Mortality Over Time

Between 2005 and 2014, the overall mortality rate doubled, increasing from 0.31% to 0.62%, representing a relative growth of 97.8% over 9 years. This corresponds to an average annual growth rate of approximately 7.88%.

Interpretation: The increasing trend, despite fluctuations such as a dip between 2007 and 2008, suggests a genuine deterioration in mortality rather than a simple effect of portfolio growth.



$$\text{Average Growth of Mortality Rate} = \left(\frac{\text{Mortality Rate of smokers 2014}}{\text{Mortality Rate of smokers 2005}} \right)^{\frac{1}{2014-2005}} - 1.$$

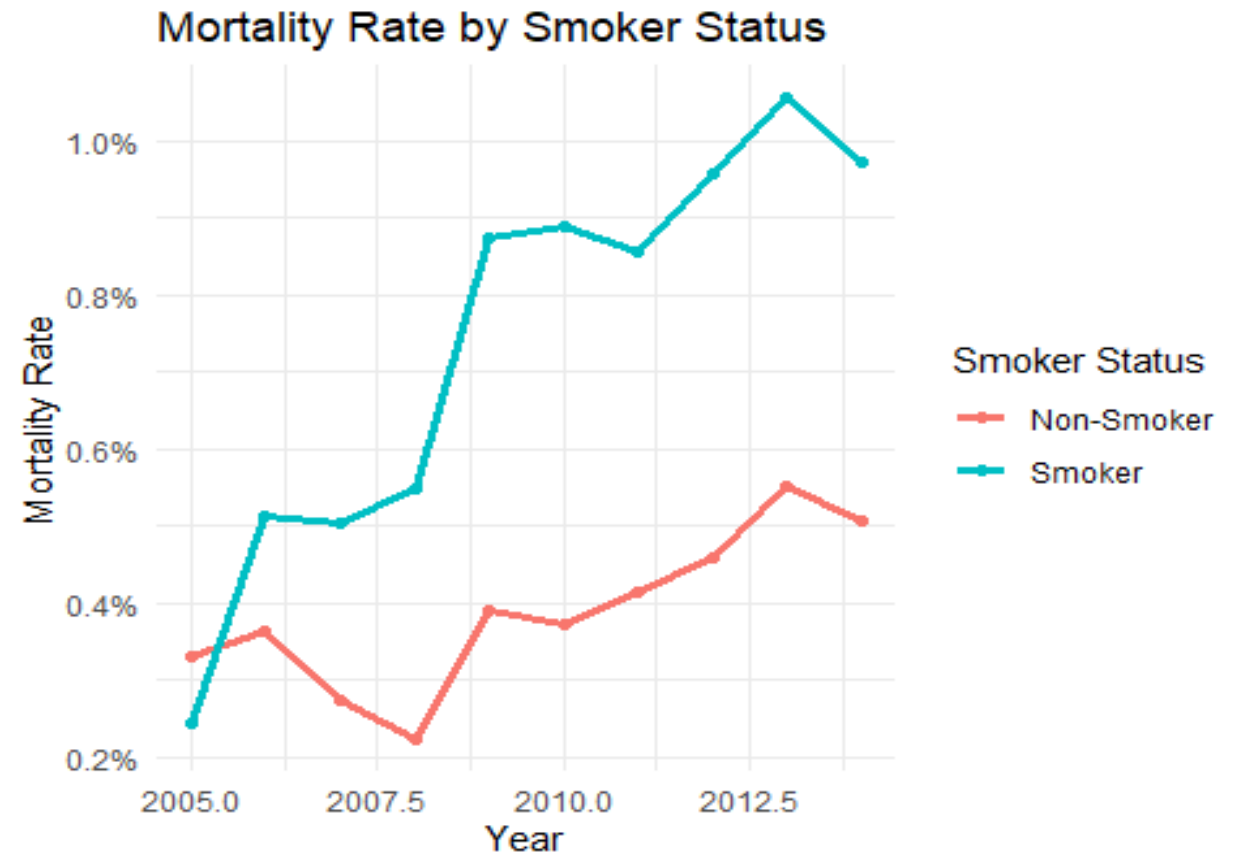
$$\text{Growth of Mortality Rate} = \left(\frac{\text{Mortality Rate of smokers 2014} - \text{Mortality Rate of smokers 2005}}{\text{Mortality Rate of smokers 2005}} \right).$$

MORTALITY ANALYSIS

Mortality Rate by Smoker Status

Mortality rates increased in both smoker and non-smoker groups. However, the increase among smokers is particularly striking and significantly outpaces that of non-smokers.

Interpretation: This strongly suggests that smokers are driving a large portion of the observed increase in total mortality and require focused underwriting attention



MORTALITY ANALYSIS

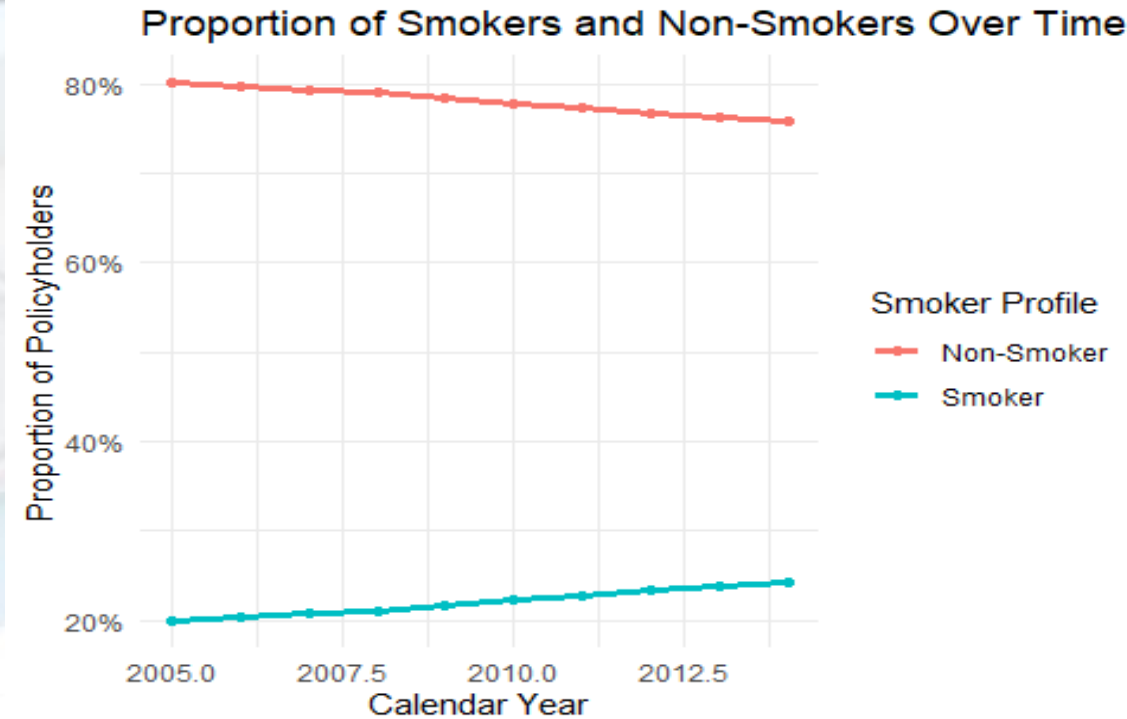
Evolution of the Smoker Composition in the Portfolio

The proportion of smokers has increased steadily in the portfolio:

- +21.68% between 2005 and 2014.
- +2.2% on average per calendar year.

Meanwhile, non-smokers declined by 5.38% in the same period (approx. -0.61% per year).

Interpretation: The growing weight of high-risk subpopulation (smokers) in the business mix is a structural concern.



Summary of Smokers Impact:

- Smoker mortality rate grew by 301.24%, with an average annual growth of 16.69%.
- Non-smoker mortality increased by 53.38%, or 4.87% per year.

MORTALITY ANALYSIS

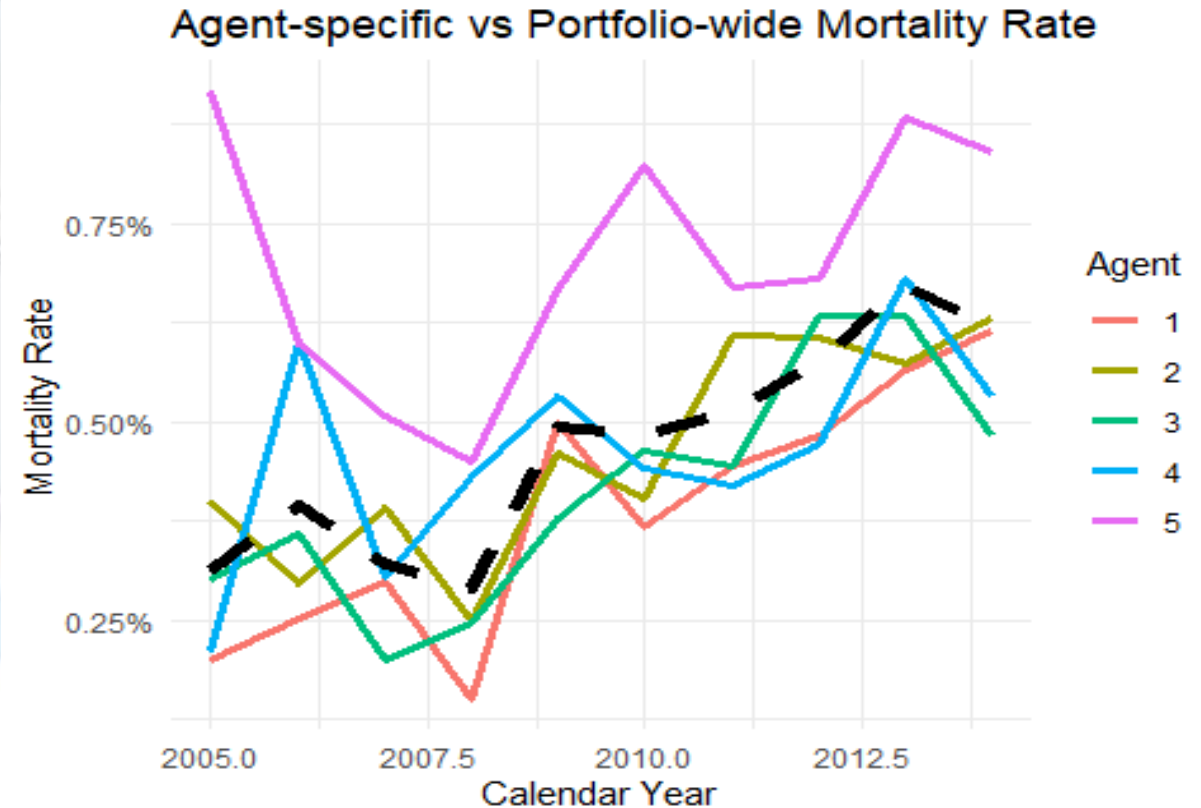
Mortality Rate by Agent

The chart below highlights the trend of mortality of each agent between 2005 and 2014.

- All agents show a general upward trend.
- Agent 5 consistently has a higher mortality rate than the rest of the portfolio.
- Greater volatility of mortality rate among Agents 1, 2, and 4 before 2010.
- Convergence around 2012–2013 indicates more similar practices or risk profiles.

Potential implications:

Further analysis of Agent 5's portfolio is warranted based on persistent above average mortality of the overall portfolio.

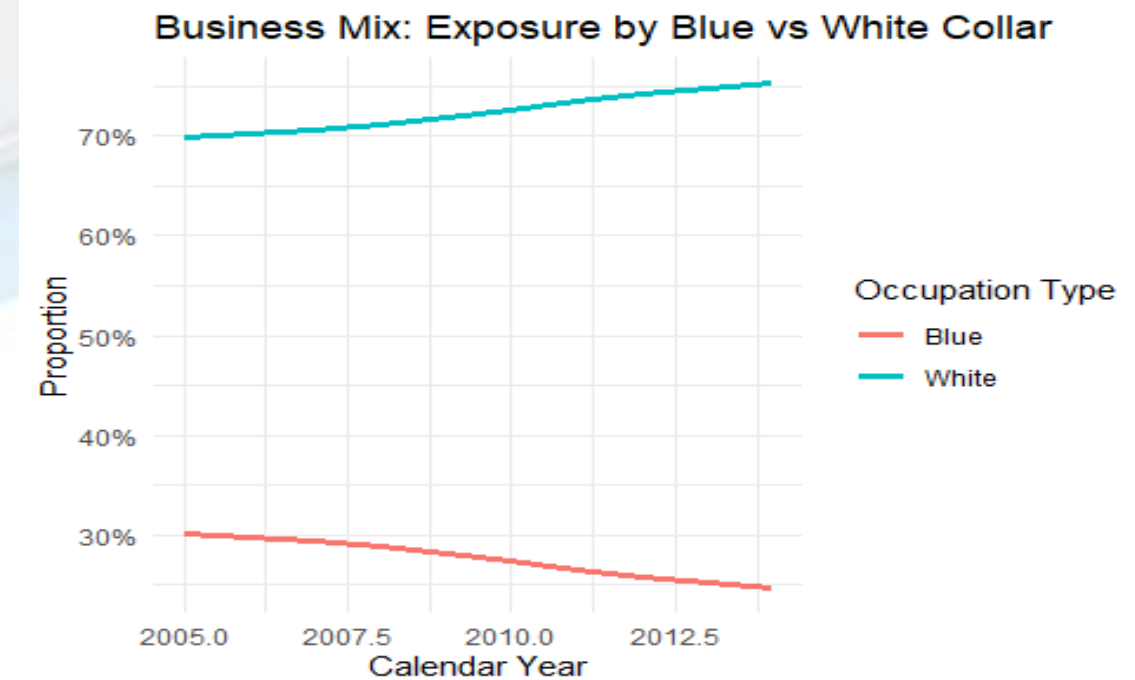
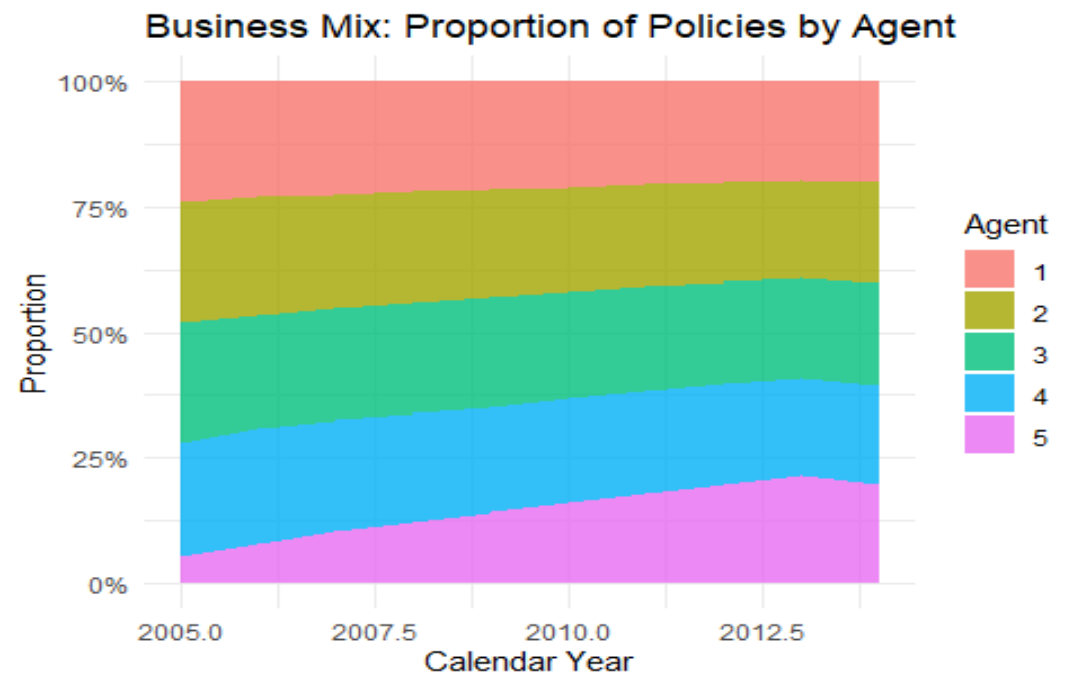


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Proportion of portfolio by Agent

Agent 5's share of the total portfolio more than doubled between 2005 and 2014, increasing from less than 10% to over 20%.

White-collar policyholders represent more than 70% of the global portfolio and seem to be increasing over the years, whereas blue-collar policyholders account for less than 30% and show a declining trend.



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Mortality by Socio-Economic Class

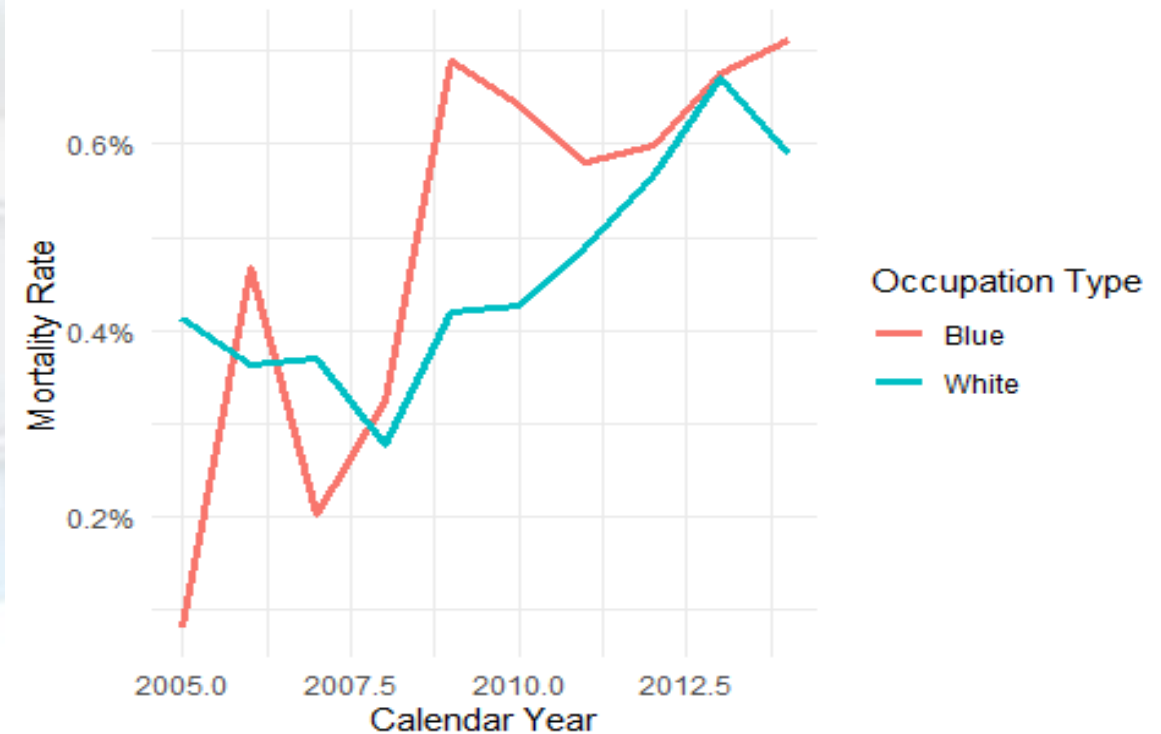
The analysis reveals a marked deterioration in mortality rates across both socio-professional segments.

However, the increase is *substantially higher and more erratic among blue-collar policyholders*, whose mortality rate rose by **789.9%** between 2005 and 2014. In contrast, the increase among white-collar policyholders was more moderate at **42.5%**.

While **white-collar individuals consistently account for over 70% of total exposure** and demonstrate lower mortality, the **relative stability of the business mix over time** indicates that the overall rise in mortality is **likely to be driven by shifts in portfolio composition**.

Furthermore, this trend points to **emerging mortality deterioration within the blue-collar subpopulation**, warranting further investigation.

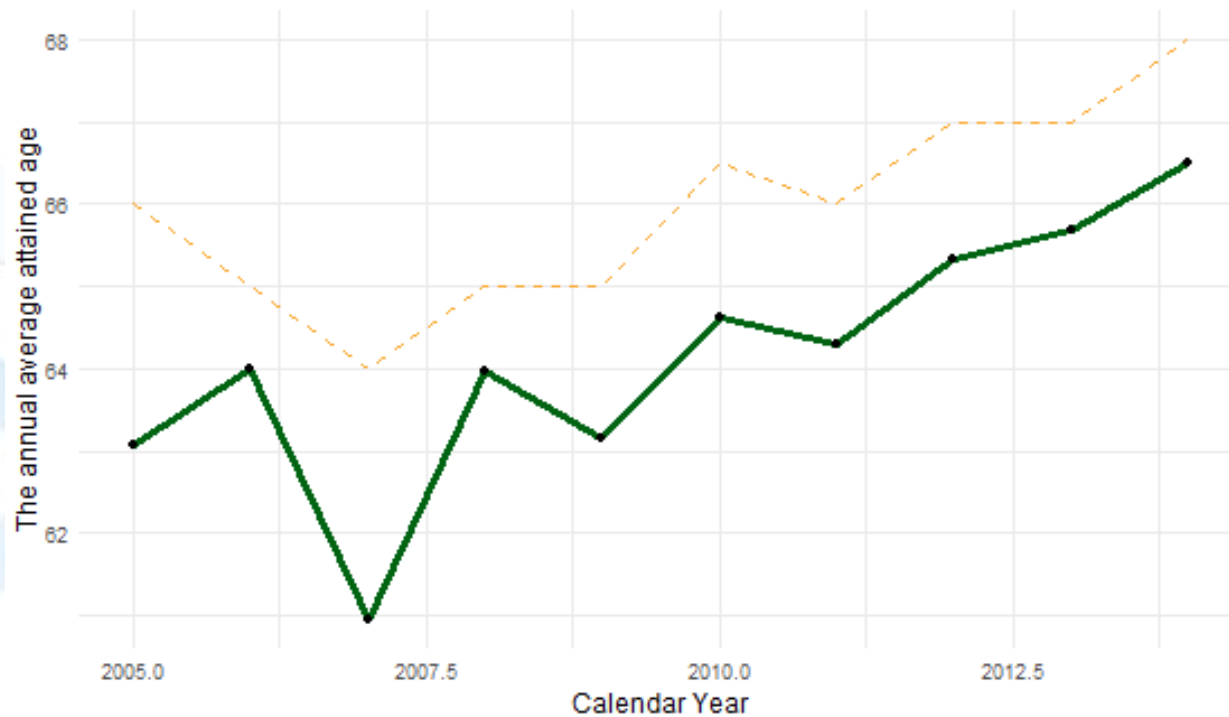
Business Mix: Blue vs White Collar



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Evolution of Attained Age at Death (Annual Averages)

The annual average and median attained age at death



Trend Implications:

Since both average and median ages at death are rising (as shown in your graph), this may suggest improving longevity over time, which could affect product design.

This mortality analysis covers a portfolio spanning the years 2005 to 2014.

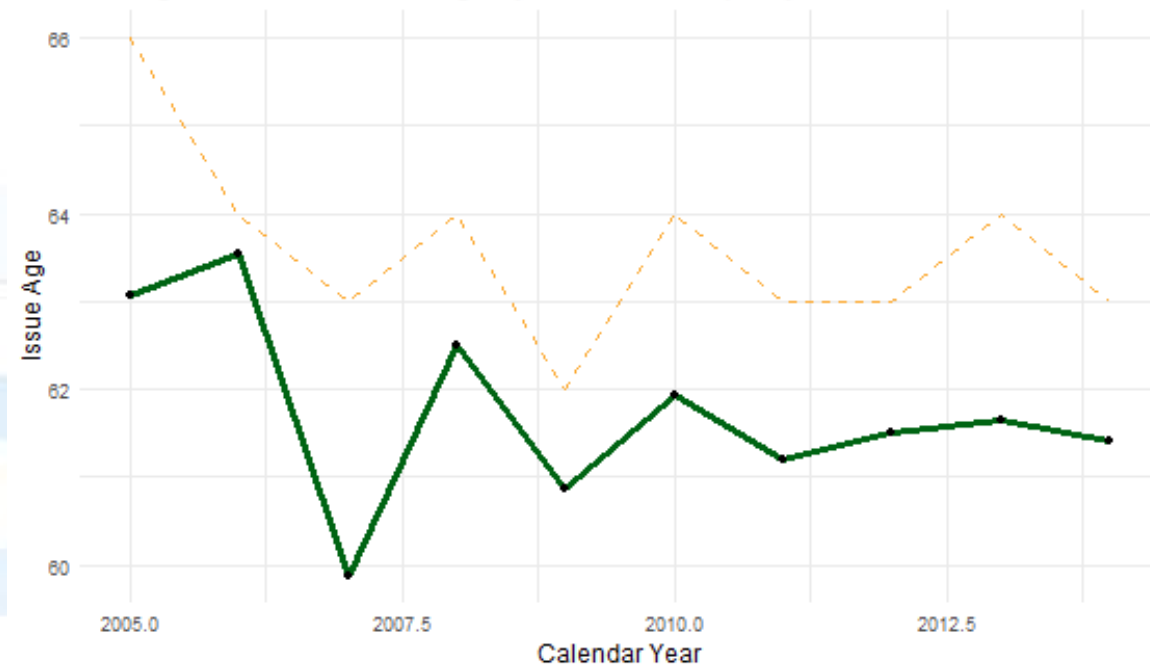
The annual average age at death across this period ranges from 60.95 to 66.50 years, with a median of 64.14 years. This suggests that 50% of policyholders are expected to pass away around the age of 64.14.

Consequently, issuing new policies to clients whose ages fall within this range requires caution. Agents should consider adjusting premiums when clients approach attained ages of 61 years or older, in order to reflect increased mortality risk.

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Evolution of Attained Age at Death (Annual Averages)

Average and Median Issue Age by Year of death policyholders

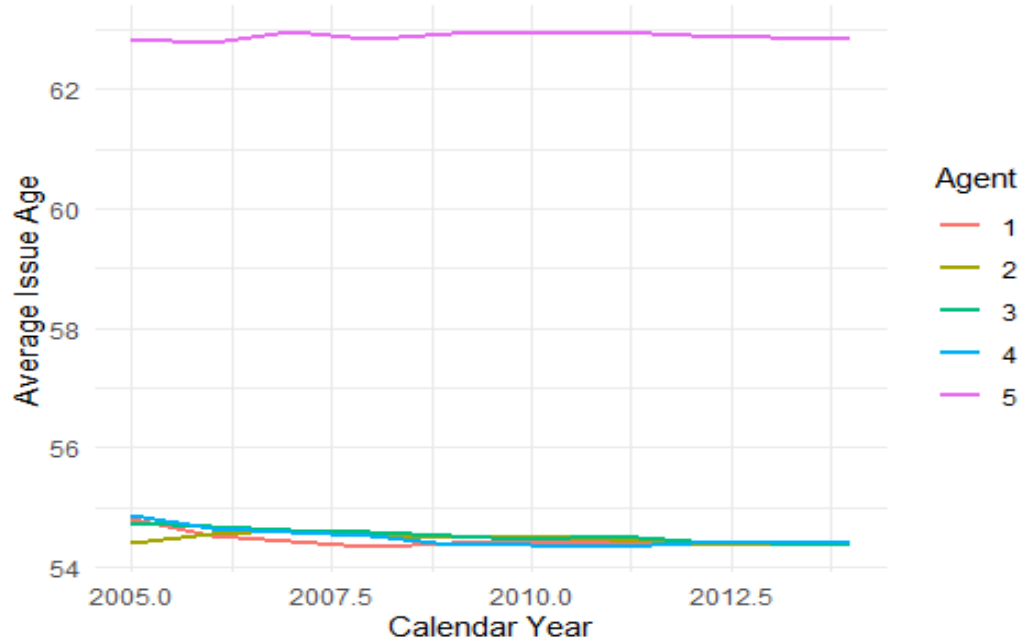


Implications:

- The **stabilization of issue ages among the deceased**, especially at higher levels, may indicate a **selection pattern that exposes the portfolio to shorter durations and higher early mortality**.
- **Underwriting and pricing** may need to be adjusted to reflect the **risk associated with late-entry insureds**, particularly when combined with low face amounts and short policy durations.

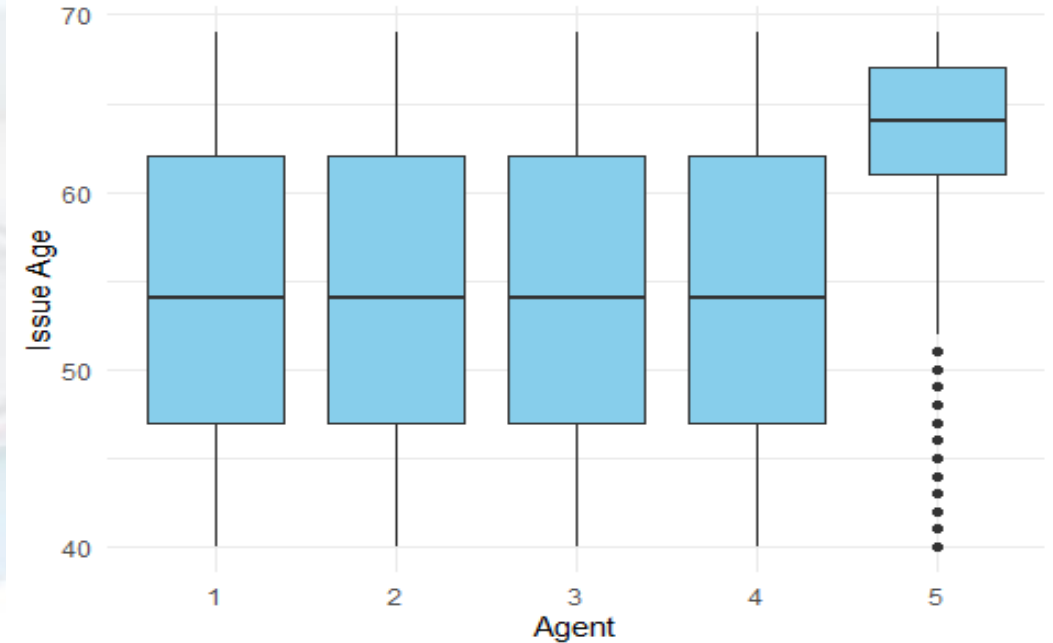
- The **average issue age** of deceased policyholders fluctuates between **60 and 64 years**, with a **notable drop in 2007**, potentially indicating the death of younger policyholders that year.
- The **median issue age** (dashed line) consistently exceeds the average in earlier years, suggesting a **right-skewed distribution** (some younger entrants dying earlier).
- From 2009 onward, both average and median values **stabilize around 63**, implying that deaths increasingly occur among policyholders who joined the portfolio at **older ages**.
- This trend reinforces prior findings: the **increase in overall mortality** may be partly driven by the **aging entry profile** of certain agents, especially Agent 5, whose clientele joins at significantly older ages.

Average Issue Age by Agent Over Time



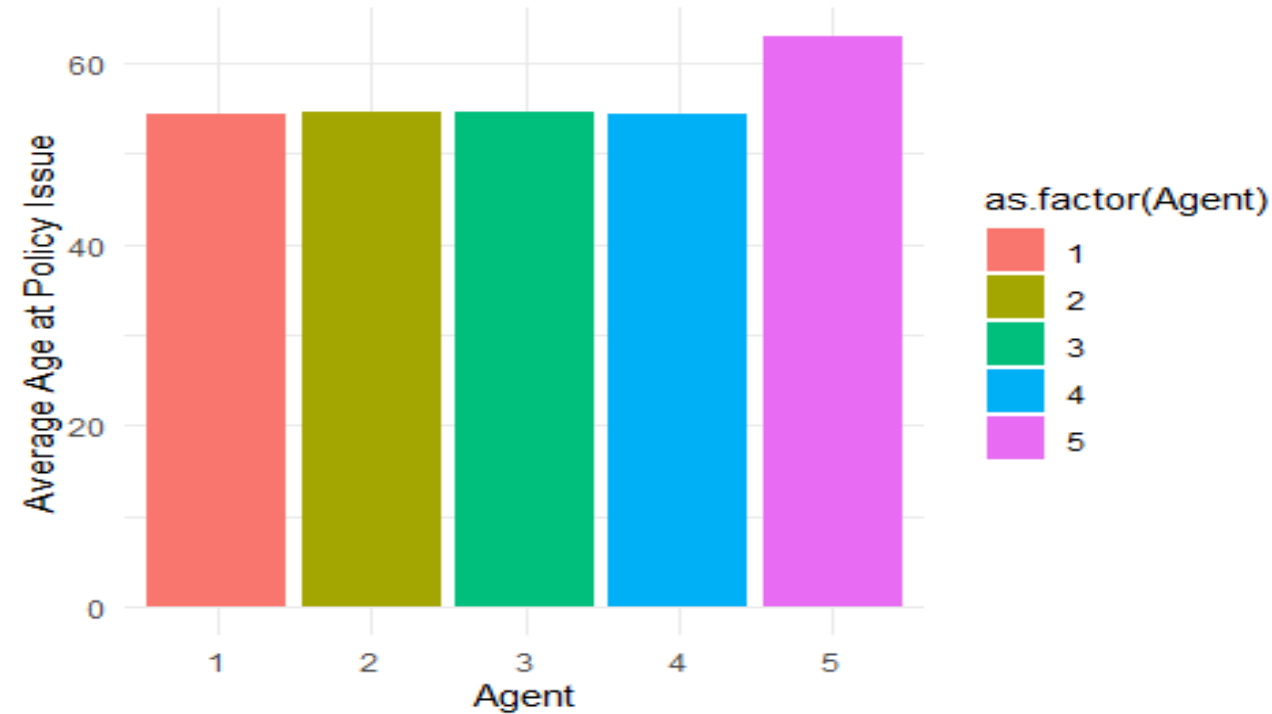
The plot below shows that the average issue ages for the first four agents remained relatively stable over time, fluctuating around 54 and 55 years old. In contrast, Agent 5 consistently exhibits significantly higher average issue ages, with values stable around 63 years old from 2005 to 2014.

Distribution of Issue Age by Agent



Moreover, while the distribution of issue ages appears more volatile and dispersed for Agents 1 to 4, Agent 5 stands out with a more concentrated and stable distribution, suggesting a different underwriting or client acquisition strategy.

Average Issue Age by Agent



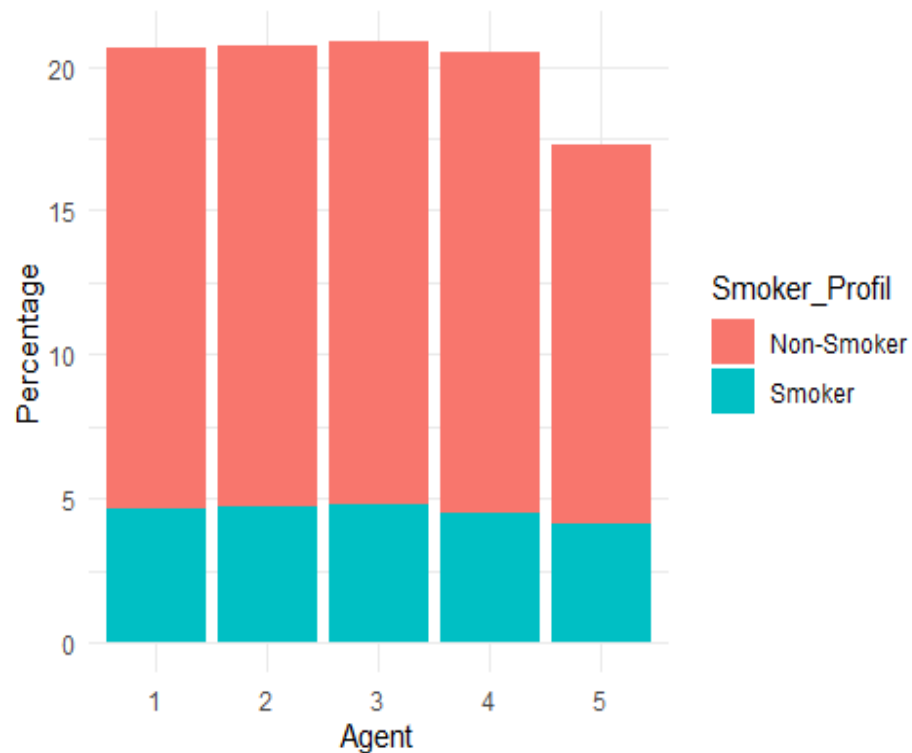
Over the period 2005 and 2014, Agent 5's clients exhibit an average issue age of 62.89 years, significantly higher than the portfolio-wide average of 56.13 years.

This persistent age gap suggests that Agent 5 tends to attract or underwrite older policyholders, which may partially explain the higher mortality rates observed in their sub-portfolio, given the direct correlation between age at issue and mortality risk.

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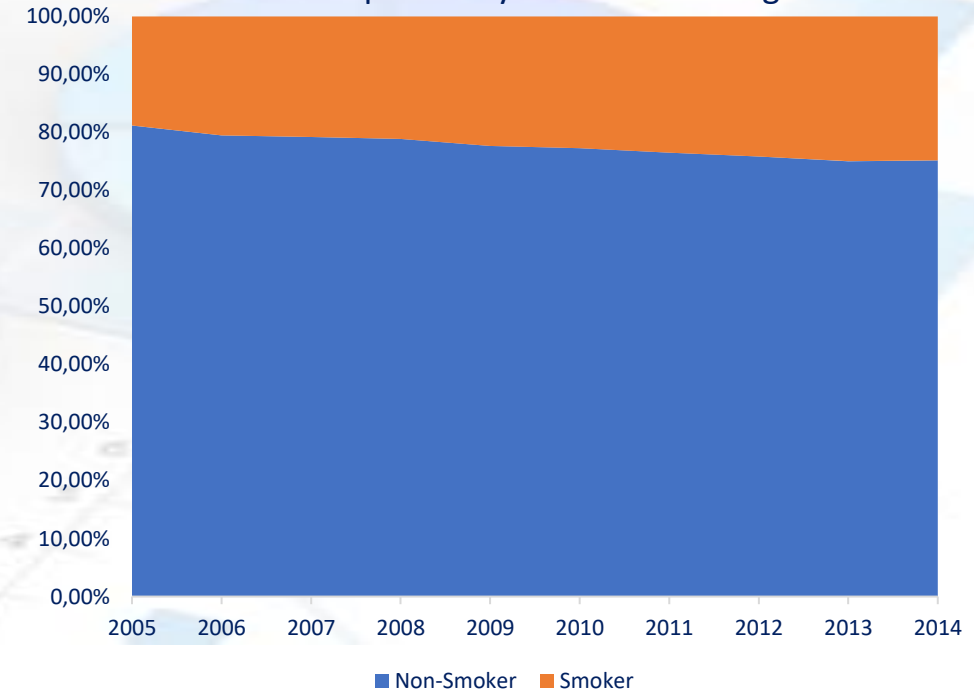
Smoker Status Distribution by Agent

Smoker Status Distribution per Agent



Agent 5 does not have a significantly higher proportion of smoker policyholders compared to other agents, although the proportion has increased over time.

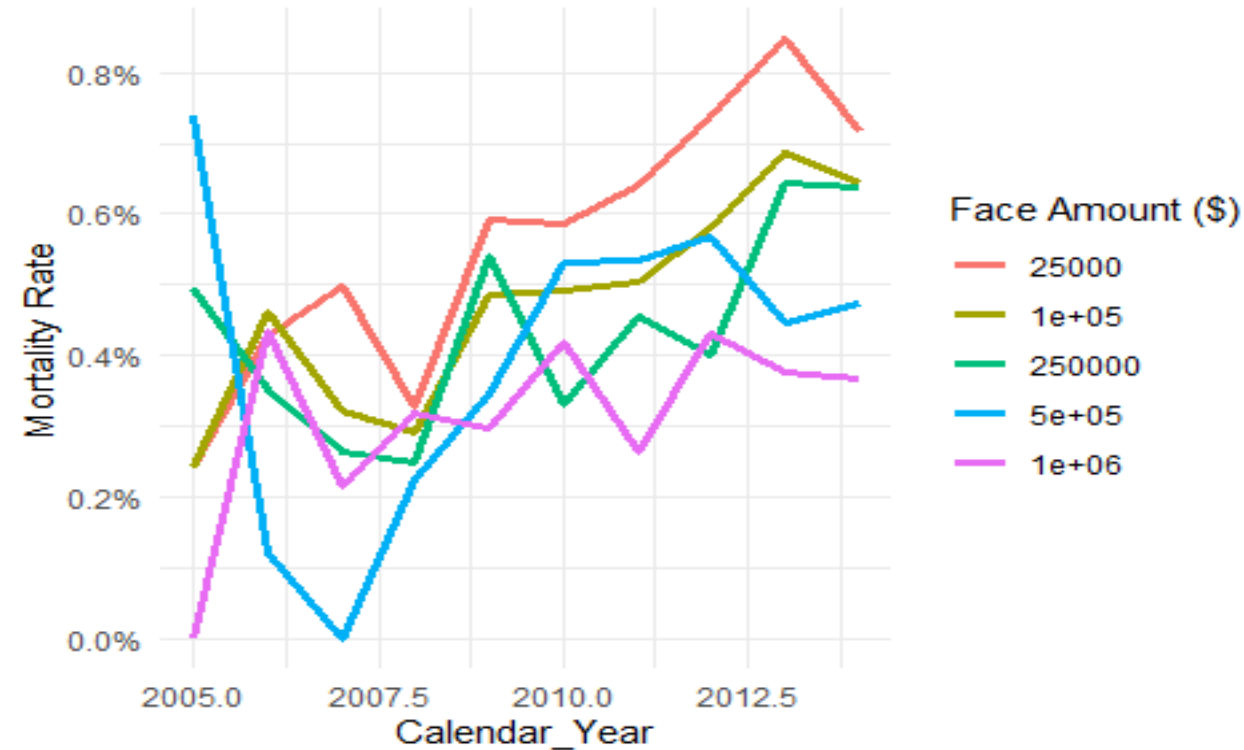
Exposure by smoker status Agent 5



However, we can partially explain the sharp increase in Agent 5's mortality rate from 2005 to 2014 by noting that his policyholders are, on average, older than those of other agents.

Additionally, the mortality rate among smoker policyholders has increased more rapidly over the years, contributing further to the overall rise in mortality.

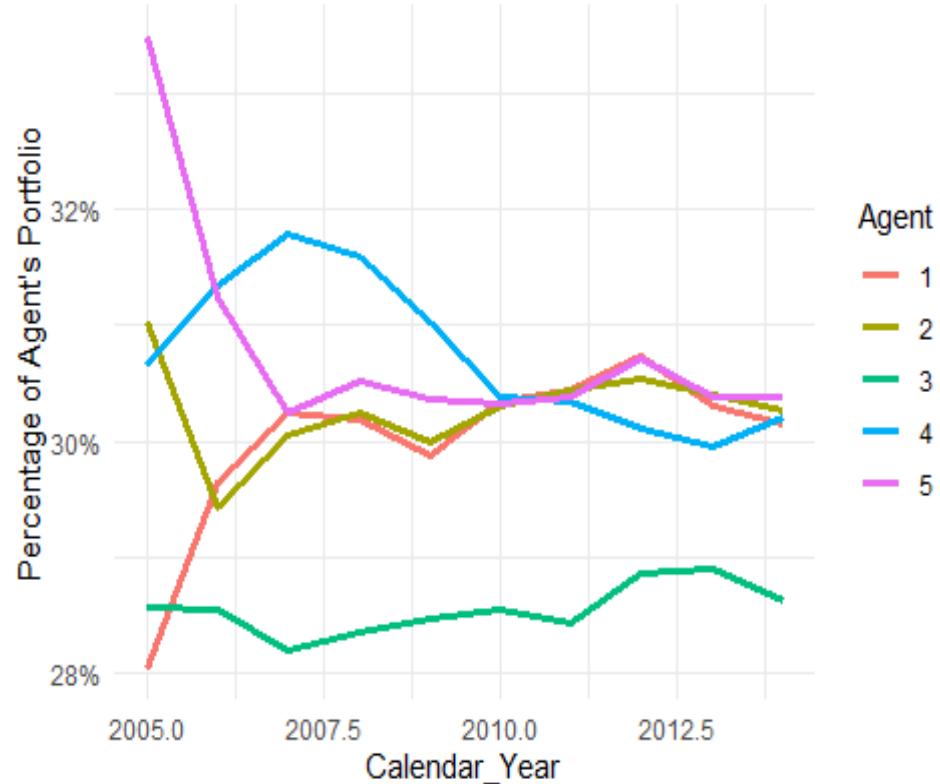
Annual Mortality Rate by Face Amount



We observe a deterioration in mortality over time across all face amount bands, except for policies with a face amount of \$500,000. For this group, there is a sharp decline in mortality between 2005 and 2007, followed by a strong upward trend in mortality rates thereafter. Mortality rates within these subgroups appear to be highly volatile.

In addition to the overall increase in annual mortality rates by face amount, the graph highlights persistent excess mortality among policies with face amounts of \$25,000 and \$100,000 throughout the observation period.

Share of \$25,000 Policies in Agent Portfolios



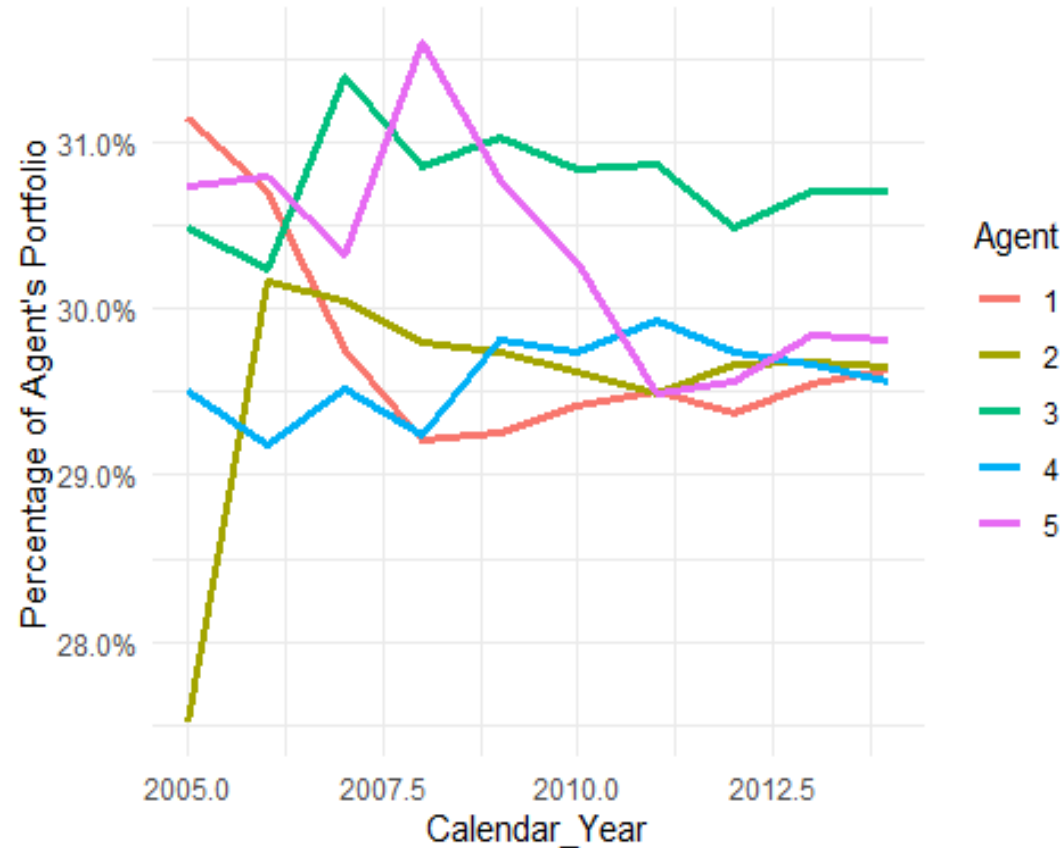
Comments:

- Policies with a face amount of \$25,000 are the most common across all agents' portfolios.
- Between 2007 and 2014, they consistently accounted for approximately 30% to 32% of each agent's portfolio. After 2006, we observe a convergence of these shares into a narrow range of 30%–31%.
- Agent 5 experienced a sharp decline in the proportion of \$25,000 policies between 2005 and 2007, dropping from over 33% to approximately 30.5%, before stabilizing slightly above 30%.
- Given that policyholders with \$25,000 face amount contracts exhibit higher mortality rates than other face amounts, the increasing and dominant presence of these contracts contributes to the rising mortality observed in individual agent portfolios.
- As a result, this also helps explain the overall increase in the portfolio-level annual mortality rate.

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Total Number of Contracts Over Time per face amount

Share of \$100,000 Policies in Agent Portfolios



Comments:

- The share of \$100,000 face amount policies ranged between 29% and 31% across all agents from 2005 to 2014.
- Toward 2014, these proportions tended to converge around 30% for most agents, with the notable exception of Agent 5, whose share of \$100,000 policies remained consistently higher, at around 31%, throughout the entire period.

These two segments (\$25,000 and \$100,000) also represent an increasing and substantial share of the agents' portfolios over time.

The combination of high mortality and growing exposure to these riskier face amount groups largely explains the upward trend in overall portfolio mortality observed from 2005 to 2014.

KEY FINDINGS

Overall Mortality Experience

The portfolio-wide mortality rate doubled, from 0.31% in 2005 to 0.62% in 2014.

This represents a 97.8% increase, or +7.88% average annual growth.

The trend reflects a genuine deterioration rather than mere portfolio expansion.

Smoker Impact

The proportion of smokers increased by +21.68%, while non-smokers declined by -5.38%.

Mortality among smokers rose by +301.24%, vs. +53.38% among non-smokers.

Smokers are a key driver of the portfolio's mortality deterioration

Face Amount Segmentation

Policies with \$25,000, \$100,000, and \$250,000 face amounts have consistently higher mortality rates. These groups have become increasingly dominant in the portfolio.

For instance, \$25,000 policies stabilized at 30 to 32% of all agents' portfolios post-2007.

The combination of higher risk and growing exposure explains much of the rise in overall mortality.

Occupational Class

Mortality increased in both socio-economic segments, but more sharply among blue-collar clients (+789.9%).

White-collar clients saw a +42.5% increase, but represented over 70% of the portfolio, a proportion that remained stable.

This suggests the rise in mortality stems from within-group deterioration more than mix shifts.

Issue Age

Between 2005 and 2014, the average issue ages for Agents 1 to 4 remained relatively stable, fluctuating around 54 to 55 years old, with a wider and more volatile distribution of ages. In contrast, Agent 5 consistently issued policies to significantly older clients, with an average issue age of 63 years and a more concentrated age distribution.

This clear divergence suggests that Agent 5 employs a distinct client acquisition or underwriting approach, focusing on an older demographic. Given the well-established link between issue age and mortality risk, this tendency likely contributes to the higher observed mortality rates within Agent 5's portfolio segment. These findings highlight the importance of considering agent-specific characteristics when analyzing portfolio risk and mortality experience.

Attained Age at Death

The median of the annual averages of attained age at death is 64.14 years, with yearly averages ranging from 60.95 to 66.50 years over the period 2005 and 2014.

This narrow age range indicates that a large proportion of claims occur within this specific interval, which could support refined pricing or underwriting adjustments, particularly from age 61 onward, where mortality risk appears to accelerate.

CONCLUSION AND RECOMMENDATION

The deterioration in portfolio mortality between 2005 and 2014 is the result of structural changes in both risk profile and business mix, rather than random fluctuations or isolated agent behavior. The increase is driven primarily by:

- A rising share of high-risk subgroups, particularly smokers and older policyholders.
- A shift toward lower face amount policies (especially \$25,000 and \$100,000), which exhibit consistently higher mortality rates.
- The growing influence of Agent 5, whose clients are older and concentrated in riskier policy bands, amplifying portfolio-level mortality.

From an actuarial standpoint, the following actions are recommended:

- Refine underwriting guidelines for high-risk groups (e.g., smokers, older insureds).
- Reassess pricing strategies for low face amount contracts.
- Implement targeted monitoring by agent to detect and mitigate adverse selection.
- Enhance portfolio segmentation to better reflect the risk gradient across face amounts and ages.

This analysis underscores the importance of granular monitoring of mortality experience across product segments, agents, and client profiles to ensure sustainable pricing and profitability.

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